9.PT-109 American patrol boat

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
   char matrix[5][5] = {
      {'M','F','H','I','K'},
      {'U','N','O','P','Q'},
      {'Z','V','W','X','Y'},
      {'E','L','A','R','G'},
      {'D','S','T','B','C'}
   };

   char text[] = "Must see you over Cadogan West. Coming at once.";
   char msg[200], a, b;
   int i, j, k = 0;
   int r1, c1, r2, c2;

   for (i = 0; text[i] != '\0'; i++)
   {
      if (isalpha(text[i]))
      {
         a = toupper(text[i]);
         if (a == 'J')
            a = 'I';
         msg[k++] = a;
      }
   }
```

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msg[k] = '\0';

for (i = 0; i < k; i += 2)
{
   if (i + 1 >= k || msg[i] == msg[i + 1])
   {
      for (j = k; j > i + 1; j--)
         msg[j] = msg[j - 1];

      msg[i + 1] = 'X';
      k++;
      msg[k] = '\0';
   }
}

printf("Encrypted message: ");

for (i = 0; i < k; i += 2)
{
   a = msg[i];
   b = msg[i + 1];

   for (j = 0; j < 25; j++)
   {
      if (matrix[j / 5][j % 5] == a)
      {
         r1 = j / 5;
         c1 = j % 5;
      }
      if (matrix[j / 5][j % 5] == b)
```

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            {
                r2 = j / 5;
                c2 = j % 5;
            }
        }

        if (r1 == r2)
        {
            printf("%c%c", matrix[r1][(c1 + 1) % 5],
                        matrix[r2][(c2 + 1) % 5]);
        }
        else if (c1 == c2)
        {
            printf("%c%c", matrix[(r1 + 1) % 5][c1],
                        matrix[(r2 + 1) % 5][c2]);
        }
        else
        {
            printf("%c%c", matrix[r1][c2],
                        matrix[r2][c1]);
        }
        printf(" ");
    }

    printf("\n");
    return 0;
}
```

Output:


C:\Users\kailash\Downloads\(
Encrypted message: UZ TB DL GZ PN NW LG TG TU ER OV LD BD UH FP ER HW QS RZ

Process exited after 0.1438 seconds with return value 0
Press any key to continue . . .